

Assignment 3

ARMA and seasonal Processes, and ARX transfer function models

Group 49

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1. Stability

1.1 Simulation of the AR(2) Process

Five realizations of length $n = 200$ were simulated from an AR(2) process with parameters $\phi_1 = -0.6$ and $\phi_2 = 0.5$, using a fixed random seed (2026). The simulated time series are shown in Figure 1.

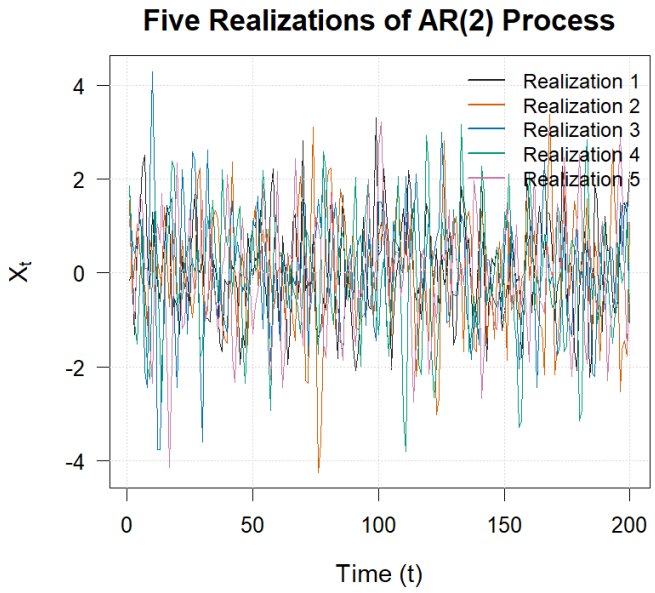


Figure 1：AR(2) Simulations (ϕ_1=−0.6, ϕ_2=0.5)

The simulated series fluctuate around zero with no apparent trend or divergence. Their variability remains stable over time, indicating that the process is stationary under the chosen parameter values.

1.2 Empirical and Theoretical ACF

The empirical autocorrelation function (ACF) was computed up to lag 30 and compared with the theoretical ACF (Figure 2).

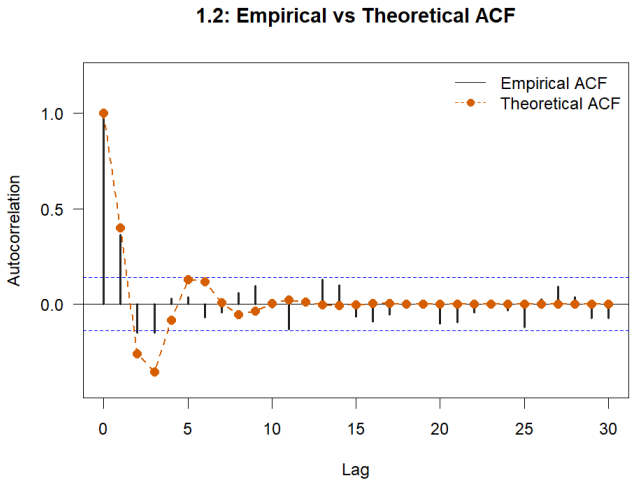


Figure 2: Empirical and theoretical ACF of the AR(2) process

The empirical ACF closely follows the theoretical ACF. Both exhibit a gradual decay toward zero, which is consistent with a stationary process. Minor deviations are observed and can be attributed to finite sample effects, especially at larger lags.

1.3 Case: ϕ1=−0.6, ϕ2=−0.3

The simulations and ACF are shown in Figures 3.

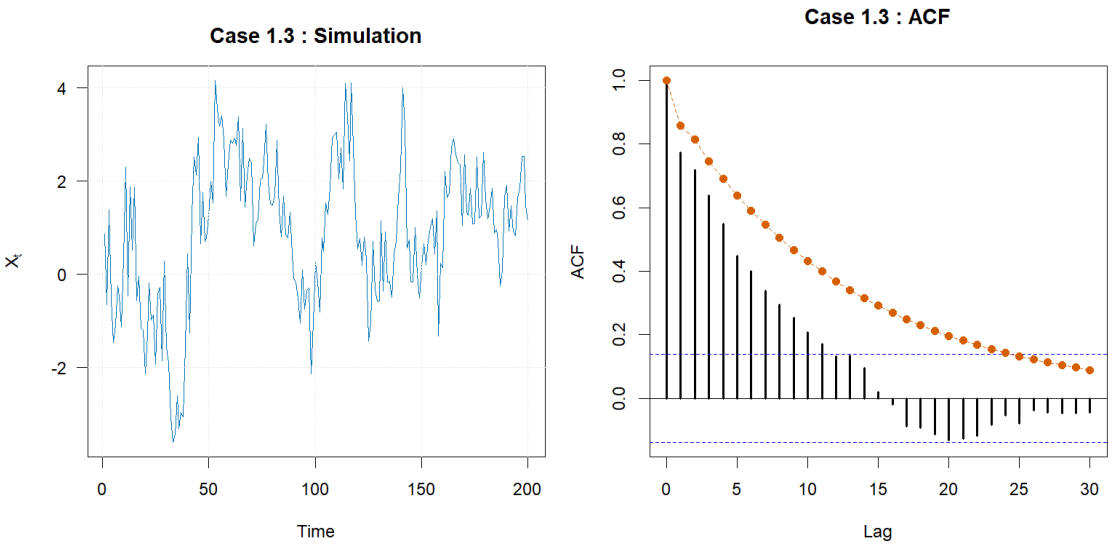


Figure 3: Simulated time series and corresponding ACF(ϕ_1=−0.6, ϕ_2=−0.3)

The ACF decays monotonically without oscillations, indicating a stable and stationary process without cyclic behavior. The simulated series remain bounded and fluctuate around zero, further confirming stationarity.

1.4 Case: ϕ1=0.6, ϕ2=−0.3

The results are shown in Figures 4.

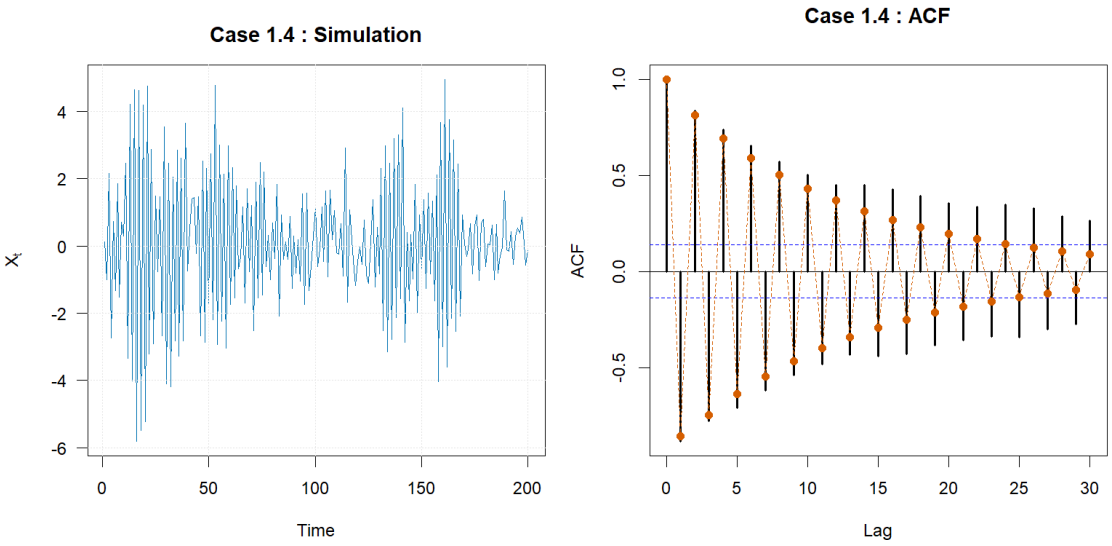


Figure 4: Simulated time series and corresponding ACF(ϕ_1=0.6, ϕ_2=−0.3)

The ACF exhibits damped oscillations, alternating between positive and negative values. This indicates a cyclic dependence structure in the series. Despite the oscillatory pattern, the ACF gradually decays toward zero, confirming that the process remains stationary.

1.5 Case: ϕ1=−0.7, ϕ2=−0.3

The results are shown in Figures 5.

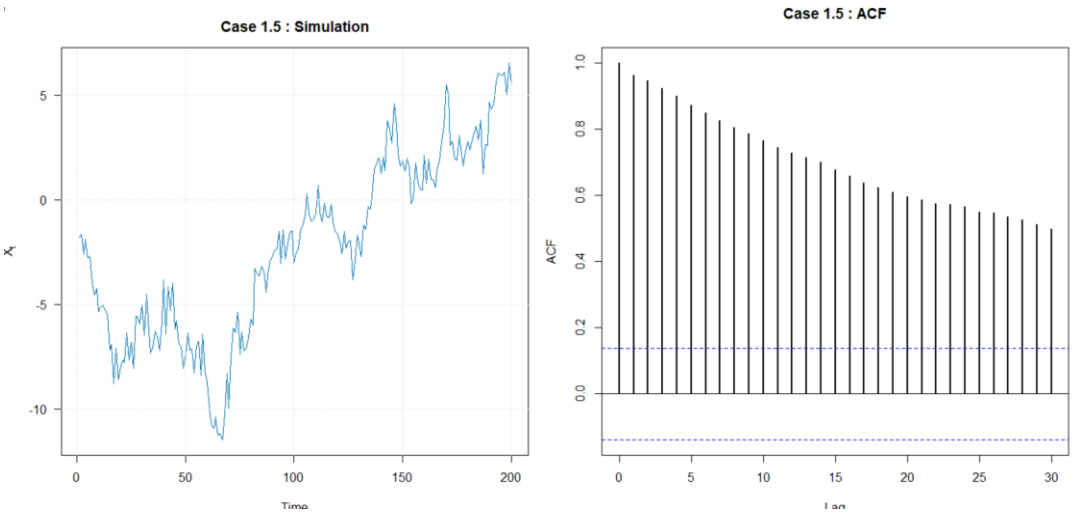


Figure 5: Simulated time series and corresponding ACF(ϕ_1=−0.7, ϕ_2=−0.3)

The ACF decays slowly, indicating strong persistence in the series. The time series shows prolonged deviations from the mean, suggesting that the process is close to the boundary of non-stationarity. Nevertheless, the series remains stationary.

1.6 Case: ϕ1=−0.75, ϕ2=−0.3

The results are shown in Figures 6.

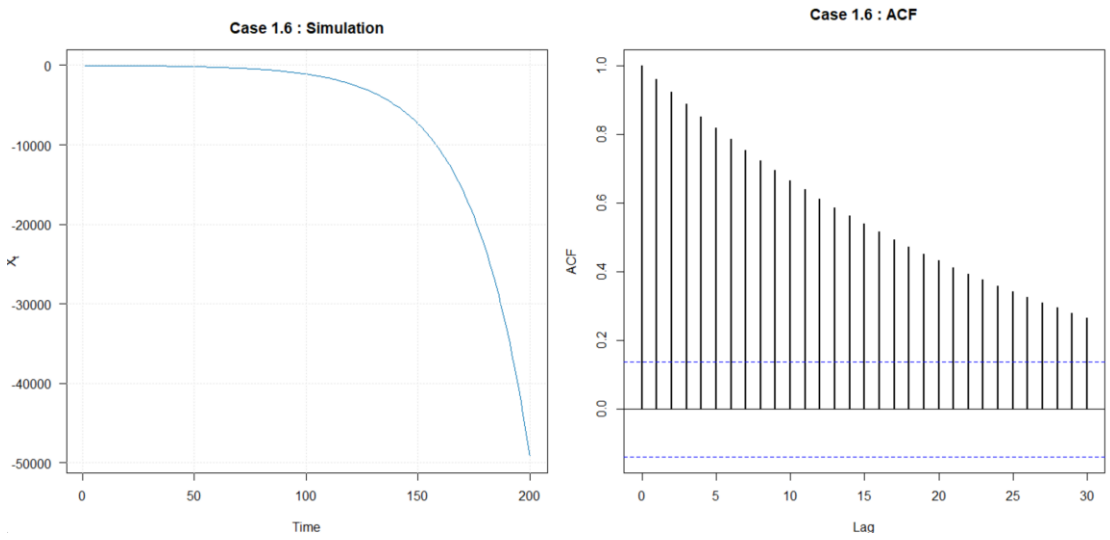


Figure 6: Simulated time series and corresponding ACF(ϕ_1=−0.75, ϕ_2=−0.3)

The time series exhibits explosive behavior, with values diverging rapidly over time. The ACF does not decay toward zero, indicating a violation of the stationarity condition. This confirms that the process is non-stationary under the chosen parameter values.

1.7 Time Series Plot vs ACF

The ACF provides a compact summary of the dependence structure of a time series and is useful for identifying persistence and oscillatory behavior. However, it is not sufficient on its own to fully assess stationarity, especially when only a limited number of lags is considered.

From the previous cases, different ACF patterns were observed. In Case 1.3, the ACF decays monotonically, indicating a stable stationary process. In Case 1.4, the ACF exhibits damped oscillations, reflecting complex roots and a cyclical dependence structure, while still remaining stationary. In Case 1.5, the ACF decays very slowly and does not appear to reach zero within the plotted lag range, suggesting a highly persistent process close to the boundary of non-stationarity. Nevertheless, the overall trend still indicates convergence toward zero, and the time series remains bounded.

In contrast, in Case 1.6, the ACF does not converge to zero and the time series shows explosive behavior, clearly indicating non-stationarity. This highlights that relying solely on the visual inspection of the ACF may be misleading, as slow decay within a limited lag range can resemble non-stationary behavior.

Therefore, a theoretical criterion is required to rigorously determine stationarity. For an AR(2) process, stationarity requires that the roots of the characteristic equation lie outside the unit circle, i.e., all roots must have modulus greater than one. Applying this condition confirms that the process in Case 1.5 is still stationary, whereas the process in Case 1.6 violates the stationarity condition.

In conclusion, both the time series plot and the ACF should be examined jointly, and theoretical conditions should be used to support the final assessment of stationarity.

2 Predicting monthly solar power

2.1 Model Validation and Residual Analysis

Let $X_t = \log(Y_t) - \mu$. The seasonal AR model can be written as:

$$(1 + \phi_1 B)(1 + \Phi_1 B^{12})X_t = \varepsilon_t$$

Expanding the model gives:

$$X_t + \phi_1 X_{t-1} + \Phi_1 X_{t-12} + \phi_1 \Phi_1 X_{t-13} = \varepsilon_t$$

Hence, the one-step-ahead prediction of X_{t+1} is:

$$\hat{X}_{t+1|t} = -\phi_1 X_t - \Phi_1 X_{t-11} - \phi_1 \Phi_1 X_{t-12}$$

and the corresponding one-step-ahead residual is:

$$\hat{\varepsilon}_{t+1|t} = X_{t+1} - \hat{X}_{t+1|t}$$

Residuals were calculated using the specified seasonal AR parameters ($\phi_1 = -0.38$, $\Phi_1 = -0.94$, $\mu = 5.72$). The residuals vs. time plot shows a stochastic pattern fluctuating around zero with no apparent trends, heteroscedasticity, or remaining seasonal structure. Furthermore, the ACF plot indicates that all autocorrelations (except lag 0) remain within the 95% confidence intervals (blue dashed lines), suggesting no significant serial correlation. These results suggest that the residuals can be reasonably approximated as i.i.d. (independent and identically distributed), implying that the model has successfully captured both the short-term and seasonal dynamics of the solar power data.

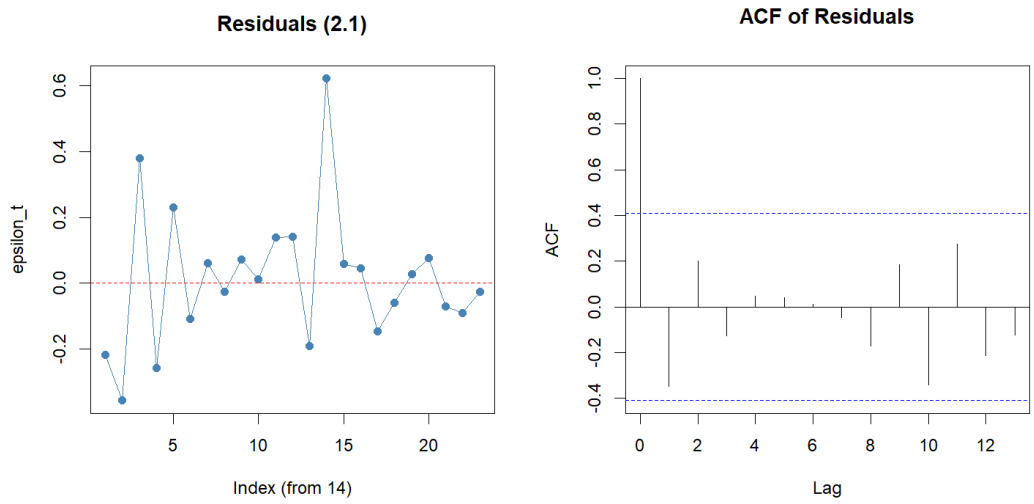


Figure 7: Model Validation of Seasonal AR Process: Time Series Plot of Residuals (Left) and Autocorrelation Function (ACF) of Residuals (Right)

2.2 Monthly Power Forecast for 2011

Based on the seasonal AR model identified, electricity generation was forecasted for the subsequent twelve months (k = 1, ..., 12 (t = 36)). The predicted values, back-transformed to the original scale (MWh), are presented in the table below and visualized in Figure 8. The forecast successfully captures the expected seasonal variations, following the historical cyclical patterns of solar power generation with peaks during the summer months.

Table1: 12-Month Solar Power Forecast for 2011 (MWh)

Time	Month Predicted Power (MWh)
Jan 2011	101.28
Feb 2011	147.23
Mar 2011	156.92
Apr 2011	142.55
May 2011	355.31
Jun 2011	538.02
Jul 2011	611.21
Aug 2011	631.89
Sep 2011	630.11
Oct 2011	508.16
Nov 2011	375.01
Dec 2011	280.49

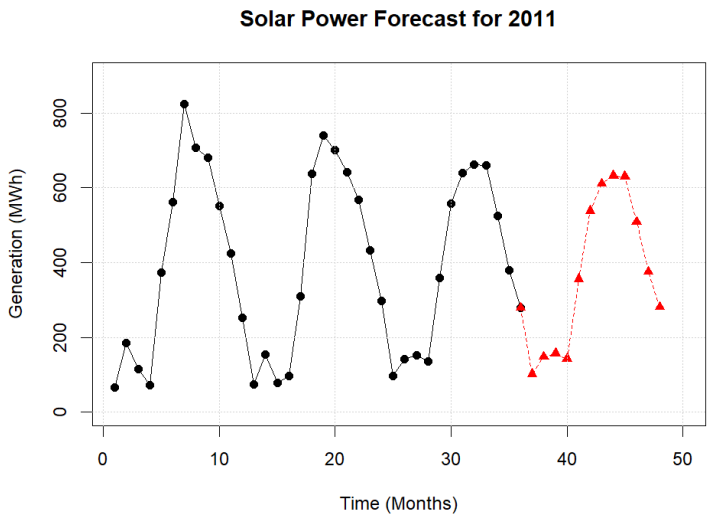


Figure 8: 12-Month Solar Power Generation Forecast for 2011 with Back-transformed Values (MWh)

2.3 Prediction Intervals and Uncertainty Analysis

The 95% prediction intervals for the twelve-month forecast horizon were calculated by simplifying the model to its AR(1) component, as per the instructions. Using the specified innovation variance $\sigma_\varepsilon^2 = 0.22^2$ and the autoregressive parameter $\theta = -\phi_1 = 0.38$, the k-step forecast error variances (σ_k^2) were derived using the formula:

$$\sigma_k^2 = \sigma_\varepsilon^2 \sum_{j=0}^{k-1} \psi_j^2$$

where $\psi_j = \theta_j$ for an AR(1) process.

The resulting confidence limits were back-transformed from the logarithmic scale to the original unit (MWh). As illustrated in Figure 9, the prediction interval correctly reflects the increasing uncertainty over time. Notably, the interval width is significantly larger during the summer peaks due to the exponential nature of the back-transformation from the log-scale, which amplifies the variance at higher power generation levels.

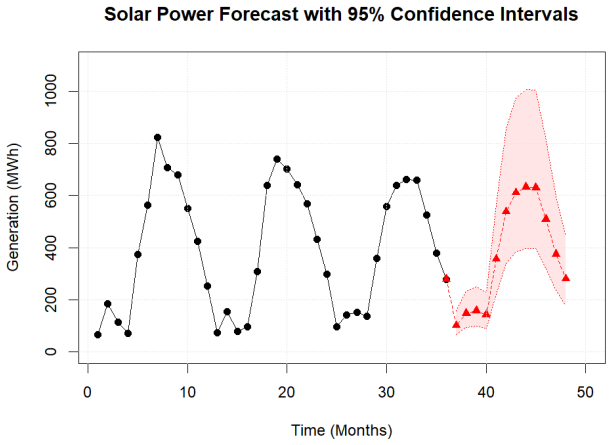


Figure 9: 12-Month Solar Power Generation Forecast for 2011 with 95% Prediction Intervals calculated based on the AR(1) simplification

2.4 Discussion on Forecast Reliability

The reliability of the forecast and the validity of the prediction intervals can be summarized as follows:

1. Forecast Trustworthiness: The predicted trend is credible, as the residual analysis in Section 2.1 confirms that the model captures the main temporal and seasonal dynamics. However, the model provides conditional mean forecasts and cannot predict random short-term variations or unexpected weather events.

2. Interval Accuracy: The prediction intervals are not fully accurate for two reasons. First, they are based only on the AR(1) component, ignoring uncertainty from the seasonal part (Φ_1). Second, the nonlinear back-transformation from the log-scale to MWh causes the intervals to widen, especially at higher generation levels, which may overestimate the true uncertainty.

Additionally, the assumption of constant variance on the log-scale may not fully reflect real-world variability.

3 An ARX model for the heating of a box

3.1 Data Visualization and Variable Dependency Analysis

The time axis is represented by the index t, corresponding to hourly observations. Although calendar time is not explicitly shown, the temporal patterns can still be interpreted in terms of daily variations and weather-driven changes. The time series plots reveal clear dynamic interdependencies between the variables. A strong negative relationship is observed between heating power (P_h) and solar radiation (G_v), as

increased solar radiation reduces the need for heating during daytime. Additionally, P_h shows a positive dependency on the temperature difference (T_{delta}), where larger differences (indicating colder external conditions) lead to increased heating demand. The solar radiation series exhibits clear daily cycles, while T_{delta} evolves more smoothly over time, reflecting slower thermal dynamics. The fluctuations in P_h indicate a responsive thermostatic control system that adjusts the heating output to maintain a constant internal temperature under varying environmental conditions. Overall, the relationships suggest that P_h is driven by both external temperature conditions and solar gains, although the exact dynamic effects and time delays cannot be fully identified from visual inspection alone.

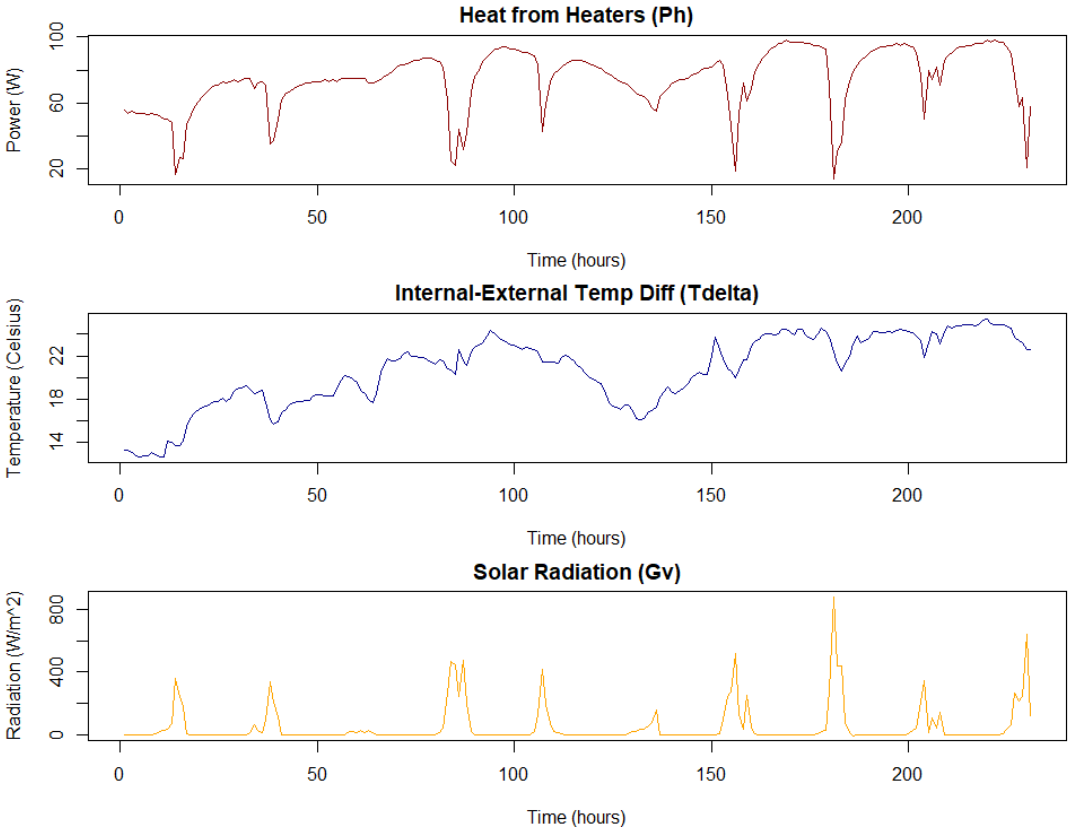


Figure 10: Hourly time series plots of heating power (P_h), internal-external temperature difference (T_{delta}), and vertical solar radiation (G_v) over 231 hours

3.2 Split the data into a train and test set

The training set consists of observations from $t = 1$ to $t = 167$, ending at the timestamp "2013-02-06 00:00". The remaining observations ($t = 168$ to $t = 231$) form the test set. From this point onward, only the training set is used for model identification and analysis, unless explicitly stated otherwise.

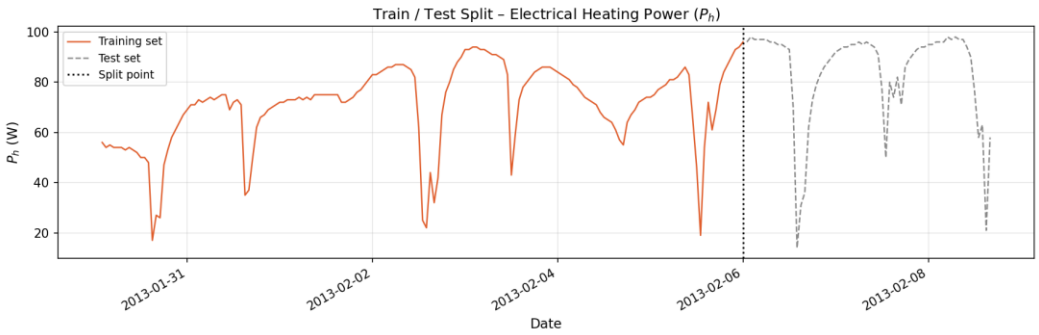


Figure 11: Train set and Test set

3.3 Investigation of Variables and Their Interrelationships

The interrelationships among the variables were investigated using the training data, revealing both static correlations and complex temporal dynamics. The scatter plots show clear linear dependencies: heating power (P_h) is positively correlated with the temperature difference (T_{delta}), indicating increased heating demand under colder external conditions, and negatively correlated with solar radiation (G_v), reflecting the compensating effect of solar gains.

The ACF of P_h exhibits a slow initial decay, indicating strong persistence due to thermal inertia in the system. A secondary rise in the ACF at higher lags suggests the presence of a diurnal cycle driven by daily temperature and solar radiation patterns.

The CCF between G_v and P_h shows a dominant negative correlation at lag 0, implying an immediate response of the heating system to solar radiation. Significant correlations at other lags are influenced by the strong autocorrelation and periodic nature of G_v, rather than indicating direct delayed causality.

While these analyses reveal the direction and temporal structure of the relationships, they cannot fully distinguish between true physical delays and correlations induced by periodic inputs. Moreover, the exact dynamic coefficients and the statistical properties of the noise term ε_t cannot be determined from these plots alone, motivating the use of formal ARX modeling in subsequent analysis.

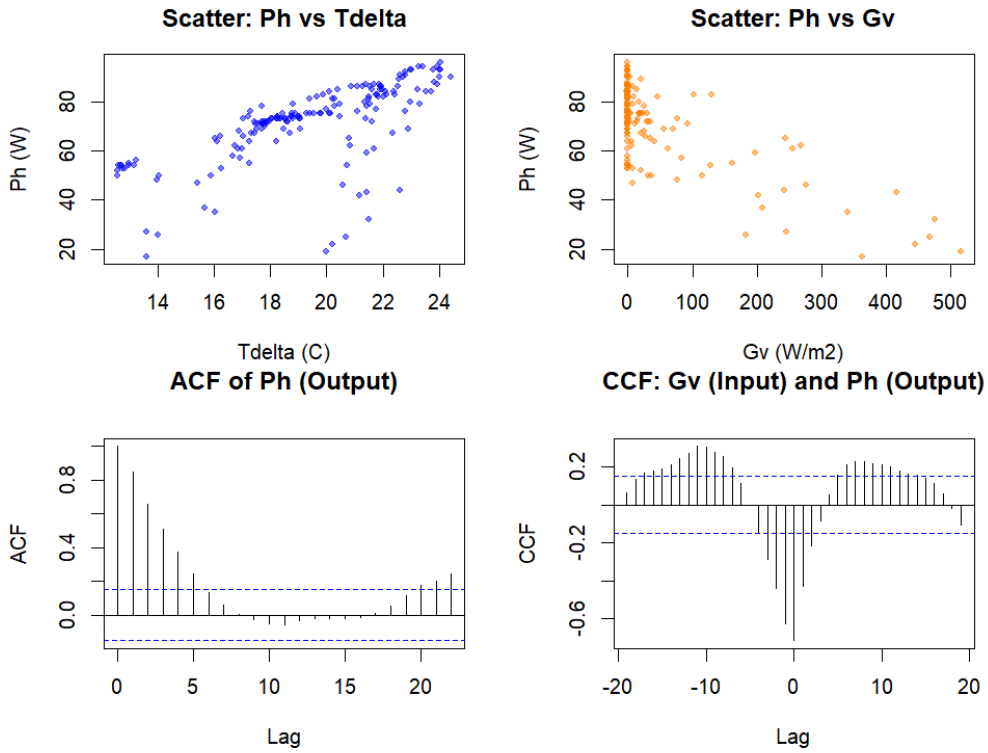


Figure 12: Exploratory data analysis of variables in the training set, including scatter plots of Ph vs. T_{delta} (top-left) and Ph vs. Gv (top-right), ACF of Ph (bottom-left), and CCF between Gv and Ph (bottom-right).

3.4 Impulse Response Estimation for Environmental Inputs

The impulse response functions (IRF) were approximated using cross-correlation analysis to quantify the dynamic impact of environmental inputs on heating power.

- Lag Selection (Lag 0 to 10): Specifically focus on the window from Lag 0 to 10 for the IRF. Although the CCF in Section 3.3 shows significant re-emerging correlations around Lag 10-12, this is interpreted as a result of the diurnal (24-hour) periodicity of the input signals rather than a direct physical causal effect. By limiting the estimation to the first 10 hours, we effectively capture the primary thermal inertia of the building while excluding the cyclic interference from the daily solar cycle.

  Findings: The IRF for T_{delta} shows an immediate positive response that decays slowly, representing the sustained heat loss compensation. The IRF for G_v displays a sharp negative impulse at Lag 0, suggesting that solar radiation immediately reduces the required heating power. The fact that the response remains significant beyond Lag 0 confirms that the box has a "memory" effect, necessitating a dynamic model.

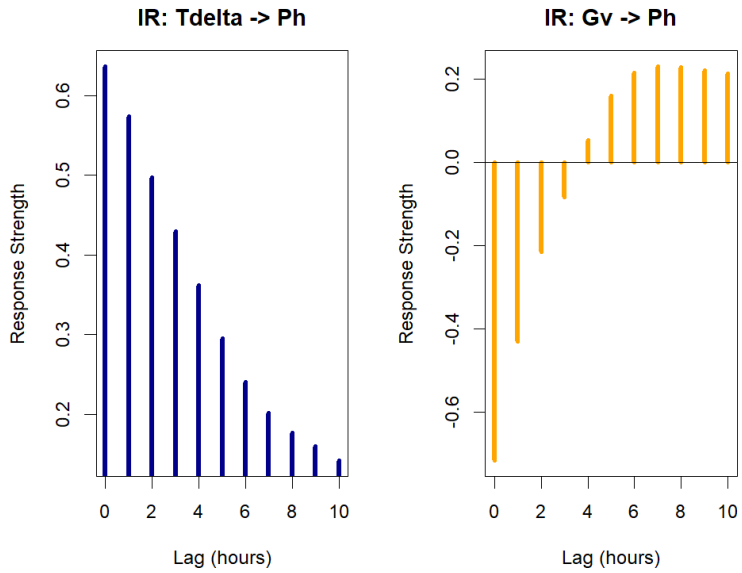


Figure 13: Estimated Impulse Response Functions (IRF) from T_{delta} (left) and G_v (right) to P_h up to Lag 10, derived from cross-correlation analysis.

3.5 Static Linear Regression and Residual Diagnostics

A static linear regression model $P_{h,t} = \omega_1 T_{delta,t} + \omega_2 G_{v,t} + \varepsilon_t$ was used as a baseline. The estimated coefficients ($\omega_1 > 0, \omega_2 < 0$) confirm that heating demand increases with temperature difference and decreases with solar radiation. The one-step-ahead predictions capture the general level but fail to reproduce short-term dynamics, indicating missing temporal dependence. The residuals show clear autocorrelation in the ACF, violating the white noise assumption. In addition, the residual CCF indicates remaining dependence on the inputs. This suggests that the static model is insufficient and that a dynamic model (e.g., ARX) is required.

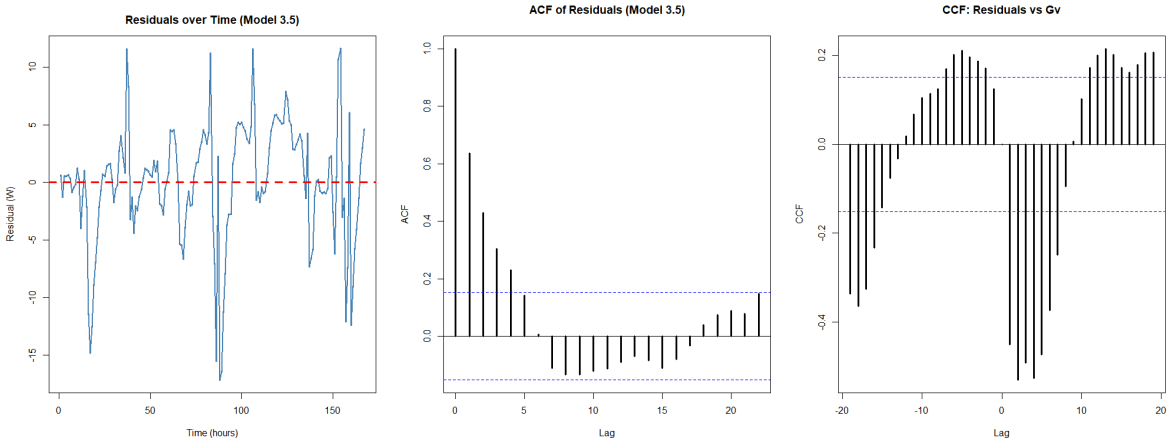


Figure 14: Residuals over time , ACF and CCF

3.6 ARX Model Estimation and Performance Evaluation

An ARX(1) model was fitted as:

$$P_{h,t} = -\phi_1 P_{h,t-1} + \omega_1 T_{delta,t} + \omega_2 G_{v,t} + \varepsilon_t$$

The one-step-ahead predictions show a clear improvement over the static model, as the predicted values follow the observed heating demand more closely compared to the static model.

The residuals are smaller in magnitude and fluctuate more randomly around zero compared to the static model. The ACF indicates that most autocorrelations fall within the confidence bounds, suggesting that the majority of the serial dependence has been removed.

The residual CCF is reduced compared to the static model, although some dependence remains, indicating that the dynamic relationship is only partially captured.

Overall, the ARX(1) model provides an improved representation of the system dynamics, mainly due to its ability to account for the thermal inertia of the system.

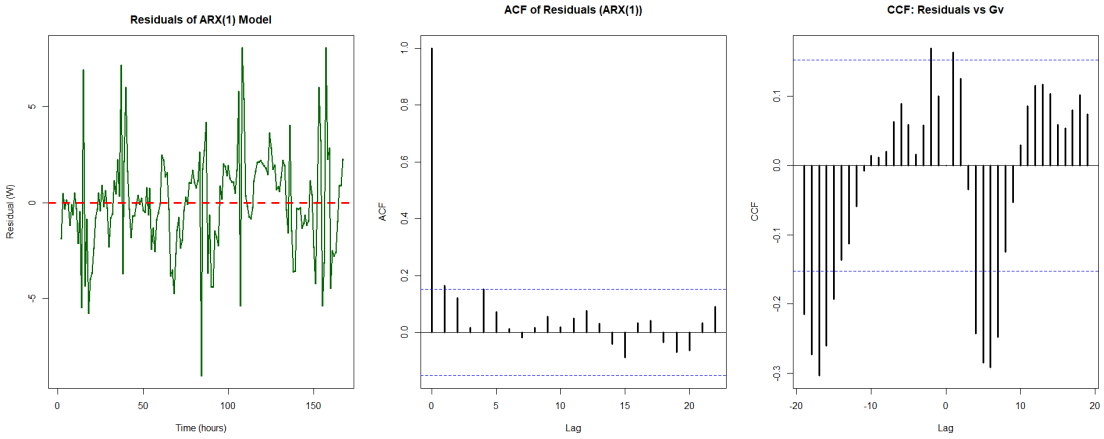


Figure 15: Residuals of ARX(1) Model, ACF and CCF

3.7 Model Order Selection using AIC and BIC

The model order was tested from 1 to 6 to cover the system's primary thermal memory. This range is chosen because 6 hours typically exceeds the thermal response time of a small test box; higher orders would risk overfitting to diurnal cycles rather than capturing internal dynamics.

Criterion Comparison: Both AIC and BIC decrease until p = 5 and rise at p = 6. This indicates that a 5-hour historical window provides the optimal balance between explanatory power and model parsimony.

Penalty Difference: BIC shows a steeper penalty for complexity than AIC (k ln(n) vs 2k), yet both identify the same minimum.

Selection: Order p = 5 is selected as the most suited, as it minimizes both information criteria and effectively filters out overparameterization.

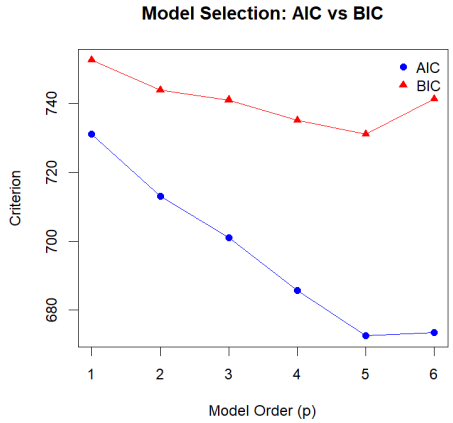


Figure 16: Comparison of AIC (blue) and BIC (red) criteria for ARX model orders p = 1 to 6

3.8 Root Mean Square Error (RMSE) Analysis

The predictive accuracy was evaluated using the Root Mean Square Error (RMSE) on the test set.

- Results: Unlike AIC/BIC which suggested p=5, the lowest RMSE is achieved at p=2. This indicates that the 2nd-order model provides the best one-step-ahead accuracy on unseen data.
- Comparison: The discrepancy suggests the 5th-order model slightly overfits the training noise. The 2nd-order model is more robust and generalizes better.

  Conclusion: Although AIC/BIC are useful benchmarks, the empirical test performance identifies Order p=2 as the most reliable structure for practical forecasting.

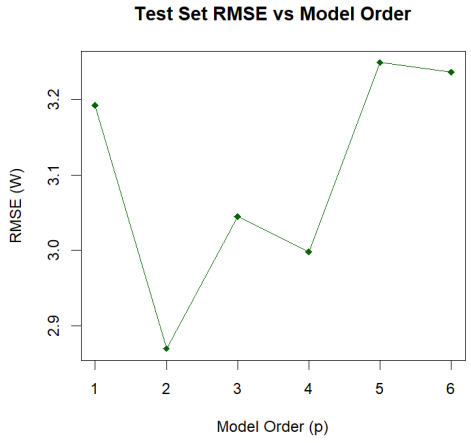


Figure 17: Root Mean Square Error (RMSE) on the test set for different model orders

3.9Multi-step Prediction (Simulation) Performance

The model was evaluated using multi-step prediction over the entire period.

In this setting, the model recursively uses its own past predictions, while the observed inputs (T_{delta} and G_v) are retained.

The simulation captures the overall pattern of the heating demand reasonably well, including the main dynamic variations. Although small errors accumulate over time, no severe drift is observed, indicating stable model behavior.

Such predictions could be applied in real-time settings, provided that reliable forecasts of the inputs are available.

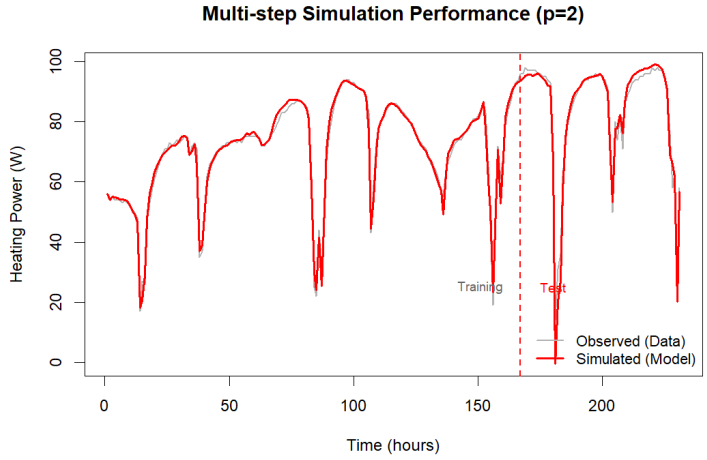


Figure 18: Multi-step simulation using the 2nd-order ARX model (Observed data is in grey; simulated output is in red)

3.10 Summarizing Conclusion

1. Evolution of Modeling: The project transitioned from a static linear model (3.5) to a dynamic ARX structure (3.6-3.9). The static model failed to account for thermal mass, resulting in autocorrelated residuals.

2. Optimal Complexity: While AIC/BIC suggested a 5th-order model, out-of-sample RMSE identified Order $p=2$ as the most robust and generalizable structure.

3. Key Drivers: Heating power is primarily driven by the temperature gradient, with solar radiation providing a significant immediate energy offset.

4. Final Verdict: The 2nd-order ARX model successfully captures the system's dynamics and is recommended for practical heat load forecasting and control applications.